PART 3 | DATA VALIDATION AND QUALITY ASSESSMENT

Data Quality Report

HealthConnect Rwanda appointment-level analytical dataset | 5,000 appointments | 30 supplied tables

Purpose and scope

This report records the integration checks, cleaning decisions, and remaining source limitations for the analytical master table. The master is one row per appointment. Source tables remain available for event-level questions and reconciliation. A zero-null result is a structural cleanliness check, not proof that every source value is factually correct.

Indicator	Observed result	Interpretation
Appointment grain	5,000 rows; 5,000 unique appointment IDs	One row per appointment after integration.
Patient coverage	4,443 distinct patients	Repeated appointments are retained in the master.
Null cells after cleaning	0	Missing source values are represented by explicit labels or documented defaults.
Financial non-negativity	No negative billed/payment totals detected	Does not confirm currency or transaction-status consistency.
Insurance identifier	Recorded across the cohort	An ID is not proof of active insurance coverage.

Cleaning approach

• Integrated the appointment-level master with one-to-one and many-to-one validations where appropriate, while preserving appointment grain.

• Standardized field types and category labels, parsed dates, and treated explicit unknown or not-recorded values as unknown rather than inventing facts.

• Applied numeric handling to measured fields and derived financial totals from linked bills and payment transactions.

• Kept the standalone Revenue_Summary separate because it has no reliable key to the appointment cohort.

Validation findings

Check	Outcome	Action or meaning
Key integrity	5,000 unique appointment keys	Passed for the appointment-grain master.
Completeness	No null cells remain in master_df_clean	Explicit unknown labels remain visible in categorical fields.
Domain checks	Age, status, and severity categories reviewed	Out-of-domain values were flagged for review rather than silently recoded.
Referential coverage	Source event tables were cohort-filtered by appointment ID	Orphan or out-of-cohort records are not treated as appointment activity.
Facility/dept mapping	4,744 / 5,000 records (94.9%) disagree across available mapping fields	Requires source-system reconciliation before location-based decisions.
Billing/payment consistency	Paid and pending statuses coexist with payments/balances	Status and transaction ledgers need a common reconciliation rule.

Important residual limitations

The cleaning pipeline eliminates technical nulls from the master but cannot resolve missing clinical history or absent measurements. In the patient cohort, chronic-condition history is not recorded for 1,131 people (25.5%). This group must remain unknown, not be treated as condition-free. Likewise, insurance coverage, bed occupancy, staffing assignments, claim settlement, and currency denomination cannot be verified from the supplied data.

Data readiness conclusion

The master is structurally usable for descriptive analysis and carefully scoped modeling. It is not yet suitable for clinical risk decisions, facility scorecards, or audited revenue reporting. Location mappings, billing status rules, and data provenance are the highest-priority remediation items.

Recommended controls

Priority	Control	Evidence to retain
1	Create authoritative DepartmentID-to-FacilityID mapping and validate it at ingestion.	Owner, mapping version, exception log, reconciliation rate.
2	Reconcile billing status against invoice and payment transactions.	Bill-level amount, payments, balance, status, and aging audit.
3	Add source lineage, currency, and event timestamps to financial data.	Source system, currency code, transaction date, refresh date.
4	Capture clinical unknown explicitly and add data-completeness indicators.	Unknown versus negative values; field-level coverage.
5	Introduce assignment and capacity events for staff, beds, and equipment.	Roster period, shift, assignment, availability, occupancy.

Scope note

This report covers the integrated master and referential checks available in the Week 6 files. It does not assert clinical validity, active insurance status, transaction settlement, or population representativeness. The EDA, inferential, and predictive findings should be read with these constraints.